

| Question Number | Answer                                                                                                                                                                                                                                                                                                                  | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(a)(i)</b>  | <p>The only correct answer is B covered in 80S (large) ribosomes</p> <p><i>A is not correct because rER is covered in 80S (large) ribosomes</i></p> <p><i>C is not correct because rER forms vesicles containing polypeptides</i></p> <p><i>D is not correct because rER forms vesicles containing polypeptides</i></p> | <b>(1)</b> |

| Question Number | Answer                                                                                                                                                                                    | Additional guidance                                          | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------|
| <b>1(a)(ii)</b> | <p>An answer that includes the following point:</p> <ul style="list-style-type: none"> <li>(smooth endoplasmic reticulum synthesises) lipid / steroids / named steroid hormone</li> </ul> | <p>ACCEPT hormones</p> <p>IGNORE (named) peptide hormone</p> | <b>(1)</b> |

| Question Number  | Answer                                                                                                                                                       | Additional guidance                                                                                                             | Mark       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>1(a)(iii)</b> | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>nucleolus (1)</li> <li>named genetic material (1)</li> </ul> | <p>ACCEPT ribosomes / rRNA</p> <p>e.g. DNA / chromatin / chromatid / chromosomes / nucleosome / mRNA</p> <p>ACCEPT histones</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                          | Additional guidance                | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------|
| <b>1(b)</b>     | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• because of water uptake / increased {volume of cytoplasm / size of vacuole} (1)</li> <li>• (increase in size due to) {synthesis/production} of {organelles / (named) proteins / enzymes} (1)</li> <li>• therefore {synthesis/production} of {new/more} cell {membrane / wall} (1)</li> </ul> | ACCEPT organelles increase in size | <b>(3)</b> |